CASE STUDY QUESTION

Q6 [Story Making] Quality Review Story Plan

Create a **Cognos Story** for the SwiftMotors Board Quality Review: Scene 1 —

- **Production Overview** (All 3 Plants)
- **Scene 2** — Defect Rate Analysis by Plant & Shift
- **Scene 3** — Top Defect Types & Root Cause Patterns
- **Scene 4** — Downtime Impact on Production Output
- **Scene 5** — Supplier Performance & Material Cost Analysis
- **Scene 6** — Quality Improvement Roadmap & Targets.

CASE STUDY

SwiftMotors Board Quality Review

Data Review Story

A comprehensive data-driven story for SwiftMotors' board meeting focusing on production performance, defect rates, root causes, downtime impact, supplier performance, and improvement targets.

Scene 1 | All Plants | Date Range | 2024-01-01 to 2024-04-30

01 Production Overview (All 3 Plants)

Provide an overview of production output, defect rates, and key performance metrics for all plants.

02 Defect Rate Analysis by Plant & Shift

Analyze the defect rates by plant and shift type (Morning, Afternoon, Night) to identify high-risk areas.

03 Top Defect Types & Root Cause Patterns

Highlight the most frequent defect types and delve into root cause analysis to identify patterns.

01 Scene 1

Production Overview (All 3 Plants)

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02 Scene 2

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03 Scene 3

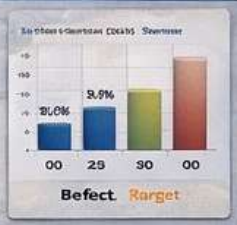
Top Defect Types & Root Cause Patterns

Highlight the most frequent defect types and delve into root cause analysis to identify patterns.

04 Scene 4

Supplier Performance & Material Cost Analysis

Analyze supplier performance, material costs and correlate scrap units with defect rates.

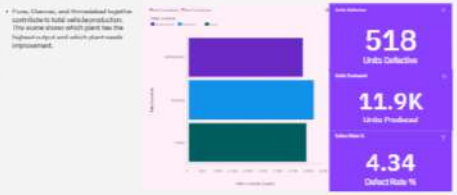


06 Scene 6 Quality Improvement Roadmap & Targets

Outline a roadmap for improving quality and set measurable targets for defect rate reduction.

Supplier	Material Cost ($)	Defect Rate	Trend
Supplier A	₹1,296.200	23.2%	↑ +0.2%
Supplier B	9203.200	23.44%	↓ 9.3%
Supplier C	9792.200	17.2%	▲ A.1%
Supplier D	9670.200	1.9.1%	B.3.1%
Supplier E	9400.200	10.96%	▼ 0.3%

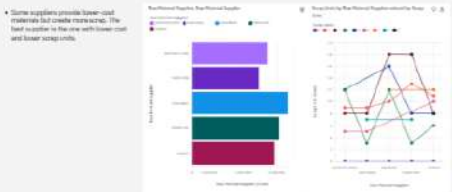
Production Overview



Defect Rate Analysis by Plant & Shift

Top Defect Types & Root Cause Patterns

Supplier Performance & Material Cost Analysis



Quality Improvement Roadmap & Targets

- SwiftMotors should focus on reducing defects, lowering downtime, improving supplier quality, and training workers. Achieving these targets will help the company reach Six Sigma quality standards.

Edit

Analytics Filters

Jump to the end of the scene to 0:10.0

Top Defect Types & Root Cause Patterns

Prev scene Next scene Scene 1 of 6 0:03.0 0:10.0

Edit

Analytics Filters

Production Overview

- Pune, Chennai, and Ahmedabad together contribute to total vehicle production. This scene shows which plant has the highest output and which plant needs improvement.

Units Defective

518

Units Defective

Units Produced

11.9K

Units Produced

Defect Rate %

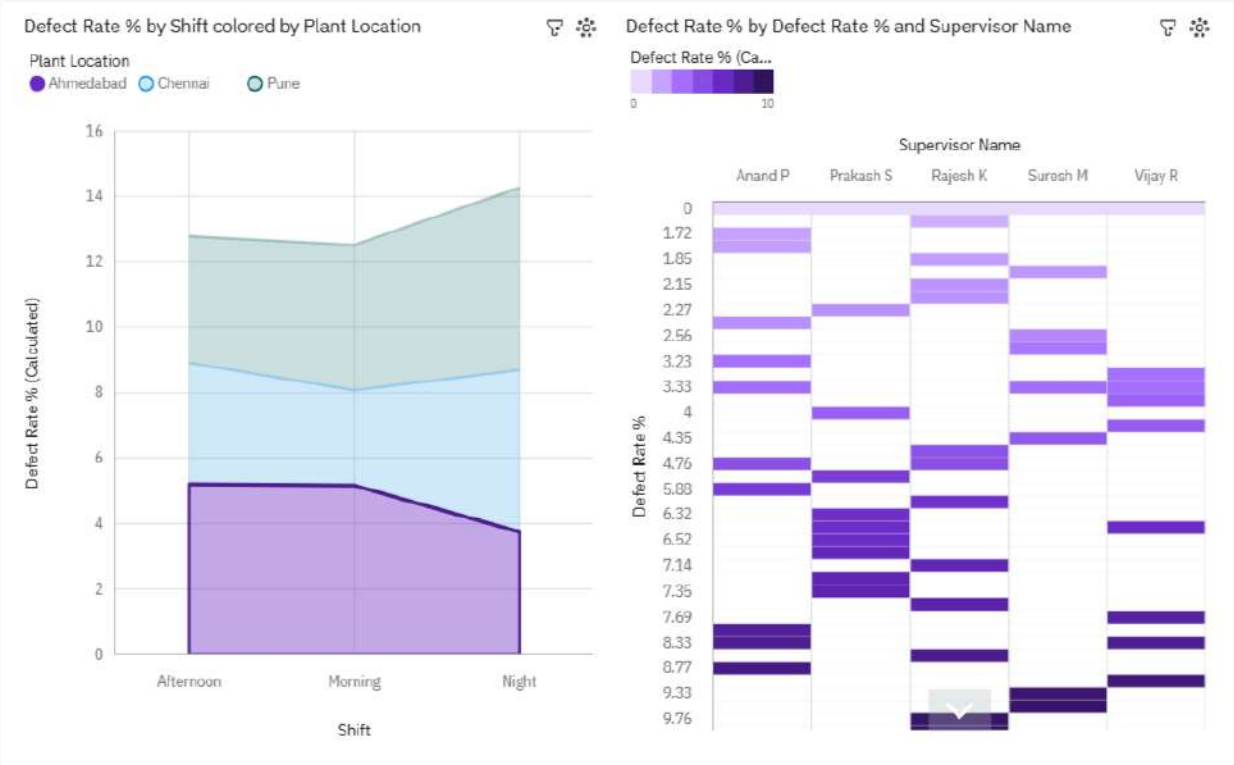
4.34

Defect Rate %

Prev scene Next scene Scene 2 of 6 0:03.5 0:05.0

Defect Rate Analysis by Plant & Shift

- Night shift has the highest defect rate in most plants. Certain supervisors and assembly lines consistently show higher defects and require closer monitoring.



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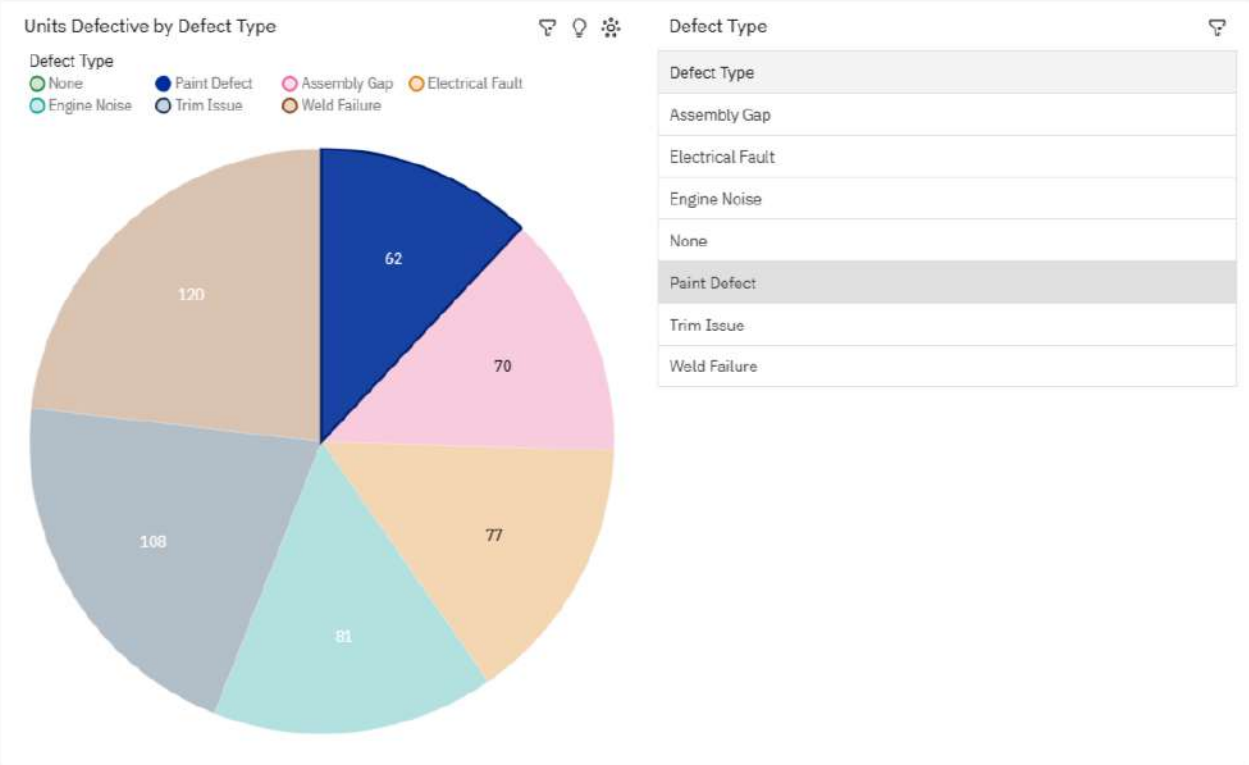
Top Defect Types & Root Cause Patterns

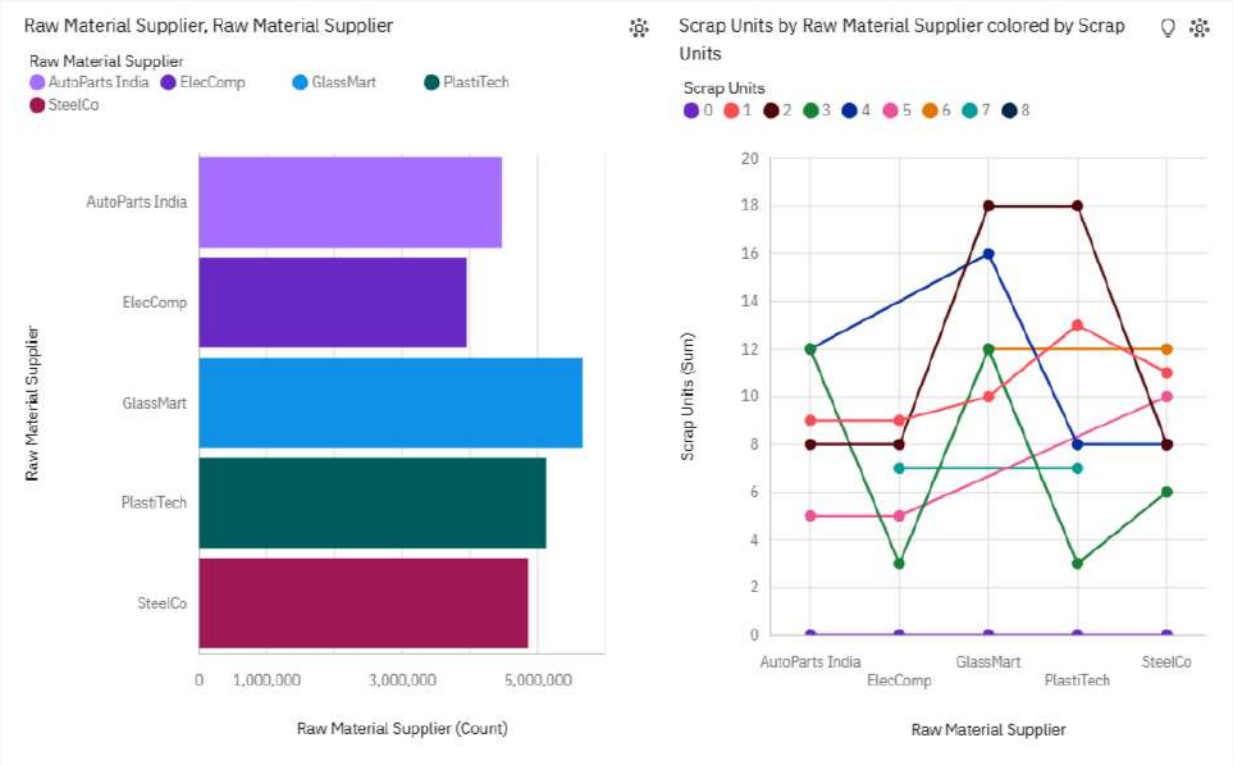
- The most common defects are mainly caused by rework issues, poor raw material quality, and assembly line mistakes. Defects that require rework increase scrap and production cost.



Prev scene
Next scene
Scene 4 of 6
0:03.7
0:05.0

Supplier Performance & Material Cost Analysis

- Some suppliers provide lower-cost materials but create more scrap. The best supplier is the one with lower cost and lower scrap units.



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Prev scene
Next scene
Scene 6 of 6
0:03.0
0:05.0

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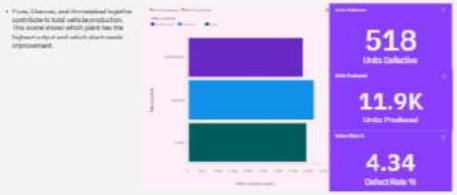
Analytics Filters

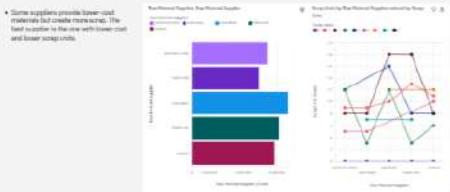
Production Overview



Defect Rate Analysis by Plant & Shift

Top Defect Types & Root Cause Patterns

Supplier Performance & Material Cost Analysis



Quality Improvement Roadmap & Targets

- SwiftMotors should focus on reducing defects, lowering downtime, improving supplier quality, and training workers. Achieving these targets will help the company reach Six Sigma quality standards.

Prev scene Next scene 6 scenes 0:00.0 0:00.0